北京交通大學

信号与系统

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Complex Frequency Domain Analysis of Continuous Time Signals and Systems

Complex Frequency Domain Analysis of Continuous Signals

- Complex Frequency Domain Representation of Continuous Signals
- Commonly used One-sided Laplace Transform
- Properties of One-sided Laplace Transform
- One-sided Inverse Laplace Transform
- Bilateral Laplace Transform and Inverse Transform

Complex Frequency Domain Analysis of Continuous LTI Systems

- Complex Frequency Domain Description of Continuous Systems
- System functions and system characteristics
- Simulation of Continuous Time Systems
- Complex Frequency Domain Analysis of System Response
- Realization of complex frequency domain analysis MATLAB

System functions for continuous-time LTI systems

- Complex Frequency Domain Description of Continuous LTI Systems
- The zero-pole distribution diagram of the system function
- System functions and system time-domain characteristics
- System function and system frequency response
- System Function and System Stability

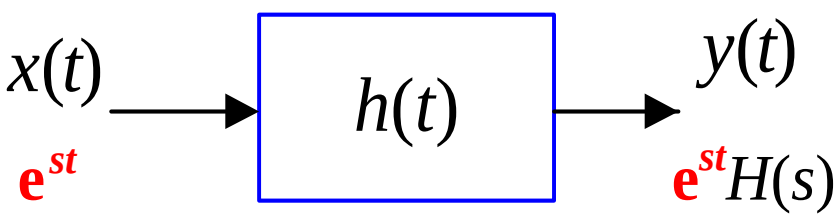
1. Complex frequency domain description of continuous-time LTI system

- Response of the complex exponential signal e^{st} through the LTI system

$$y_{zs}(t) = e^{st} * h(t) = \int_{-\infty}^{\infty} e^{s(t-\tau)} h(\tau) d\tau$$

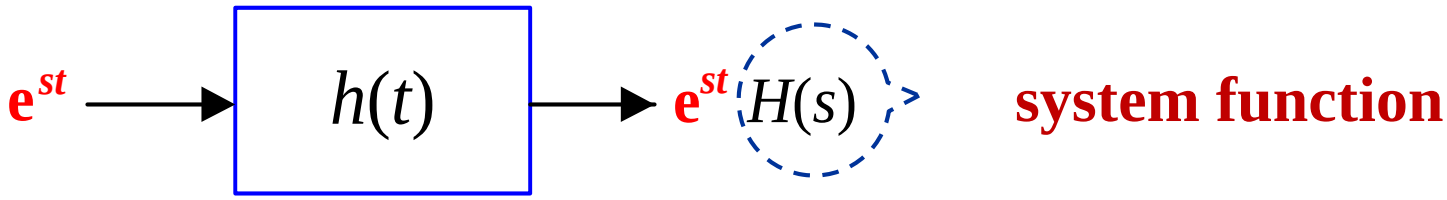
$$= e^{st} \underbrace{\int_{-\infty}^{\infty} e^{-s\tau} h(\tau) d\tau}_{H(s)}$$

$$= e^{st} H(s)$$



1. Frequency-domain description of continuous-time LTI systems

- **System function $H(s)$**



- Relationship between system function $H(s)$ and impulse response $h(t)$

$$H(s)=\mathscr{L}[h(t)]=\int_{-\infty}^{\infty} h(t)\mathrm{e}^{-st}\mathrm{d}t$$

1. Frequency-domain description of continuous-time LTI systems

- **Another Definition of System Function *H*(*s*)**

Frequency Domain Representation of Zero State Responses

$$y_{zs}(t) = x(t) * h(t)$$

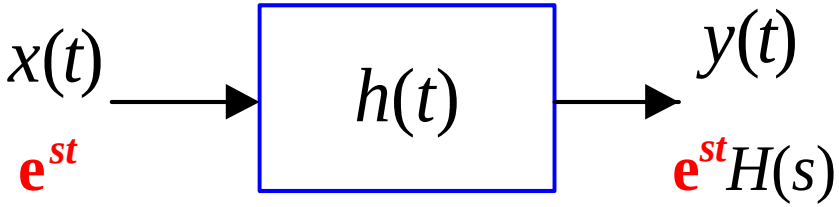
Using Laplace transform
The ***convolution characteristics*** of

$$Y_{zs}(s) = X(s)H(s)$$

$$H(s) = \frac{Y_{zs}(s)}{X(s)}$$

1. Frequency-domain description of continuous-time LTI systems

- *the physical meaning of* ***H(s)***



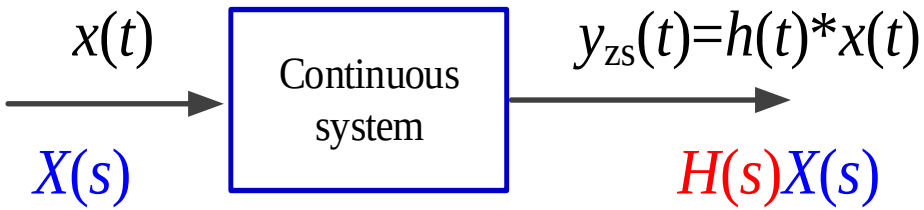
Response of a continuous LTI system through a complex exponential signal e^{st} of complex frequency s

It is still a complex exponential signal with the same frequency. The change in signal is determined by ***H(s)*** .

H(s) reflects the **transmission characteristics** of the continuous system to complex exponential signals of different frequencies.

1. Complex Frequency-domain description of continuous-time LTI systems

➢ The physical meaning of $H(s)$



The LTI system changes the input signal complex spectrum $X(s)$ into $H(s)\,X(s)$,

The law of change is completely determined by $H(s)$.

$H(s)$ reflects the transmission characteristics of the system to different complex frequency components of the input signal.

1. Complex Frequency-domain description of continuous-time LTI systems

- **The Main Methods of Solving the System Function *H*(*s*)**

① Solve from the impulse response of the system: $H(s)=\mathbf{L}[h(t)]$

② Calculated from the input-output of the system:

$$H(s)=\frac{\mathscr{L}[y_{zs}(t)]}{\mathscr{L}[x(t)]}$$

③ Calculated by the differential equation describing the continuous time system

Example: Try to find the system function $H(s)$ of ideal integrator and ideal differentiator

Solution: (1) The input-output relationship of the ideal integrator is

$$y(t) = \int_{-\infty}^{t} x(\tau)\mathrm{d}\tau$$

The impulse response of the system is

$$h(t) = \delta^{(-1)}(t) = u(t)$$

Therefore, the system function of the ideal integrator is

$$H(s) = \mathscr{L}[h(t)] = \frac{1}{s}$$

Example: Try to find the system function $H(s)$ of ideal integrator and ideal differentiator.

Solution: (2) The input-output relationship of the ideal differentiator is

$$y(t) = \frac{\mathrm{d}x(t)}{\mathrm{d}t}$$

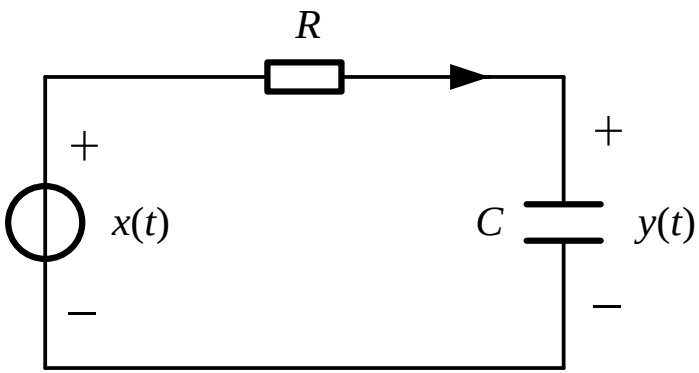
The impulse response of the system is

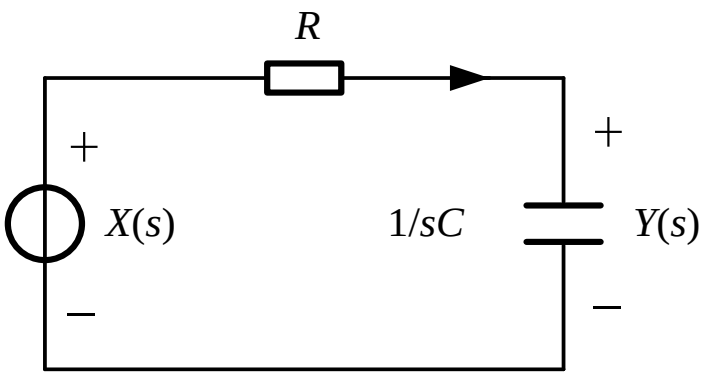
$$h(t) = \frac{\mathrm{d}\delta(t)}{\mathrm{d}t}$$

Therefore, the system function of the ideal integrator is

$$H(s) = \mathscr{L}[h(t)] = s$$

Example: In the *RC* circuit system shown in the figure, the excitation voltage source is $x(t)$, the output voltage $y(t)$ is the voltage $v_C(t)$ across the capacitor, and the initial state of the circuit is zero. Find the system function $H(s)$ and impulse response $h(t)$ of the system.





Solution: The complex frequency domain model of the RC circuit is shown in the figure. From the basic principle of the circuit, we can get

$$H(s)=\frac{Y(s)}{X(s)}=\frac{1/sC}{R+\dfrac{1}{sC}}=\frac{1/RC}{s+1/RC}$$

By Laplace inverse transformation, the impulse response $h(t)$ of the *RC* system can be obtained as

$$h(t)=\frac{1}{RC}\mathrm{e}^{-(1/RC)t}u(t)$$

Example: It is known that the differential equation describing a continuous-time LTI system is $y''(t)+3y'(t)+2y(t)=3x(t)+2x'(t) \quad t \geq 0$

Find the system function $H(s)$ and unit impulse response $h(t)$ of the system.

Solution: Apply Laplace transform on both sides of the differential equation to get

$$(s^2+3s+2)Y_{zs}(s)=(2s+3)X(s)$$

According to the definition of system function, we can get

$$H(s)=\frac{Y_{zs}(s)}{X(s)}=\frac{2s+3}{s^2+3s+2}=\frac{1}{s+1}+\frac{1}{s+2}$$

Carrying out Laplace inverse transformation, we can get

$$h(t)=(\mathrm{e}^{-t}+\mathrm{e}^{-2t})u(t)$$

1. Frequency-domain description of continuous-time LTI systems

- **The general form of the system function *H*(*s*) of continuous-time LTI systems**

If the differential equation describing the continuous LTI system is

$$a_n y^{(n)}(t) + a_{n-1} y^{(n-1)}(t) + \cdots + a_1 y'(t) + a_0 y(t) =$$
$$b_m x^{(m)}(t) + b_{m-1} x^{(m-1)}(t) + \cdots + b_1 x'(t) + b_0 x(t)$$

Using the differential characteristics of Laplace transform, the complex frequency domain equation describing the system can be obtained

$$[a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0] Y_{zs}(s) =$$
$$[b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0] X(s)$$

According to the definition of the system function of the continuous system, we can get

$$H(s) = \frac{Y_{zs}(s)}{X(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}$$

2. Zero-pole distribution diagram of the system function

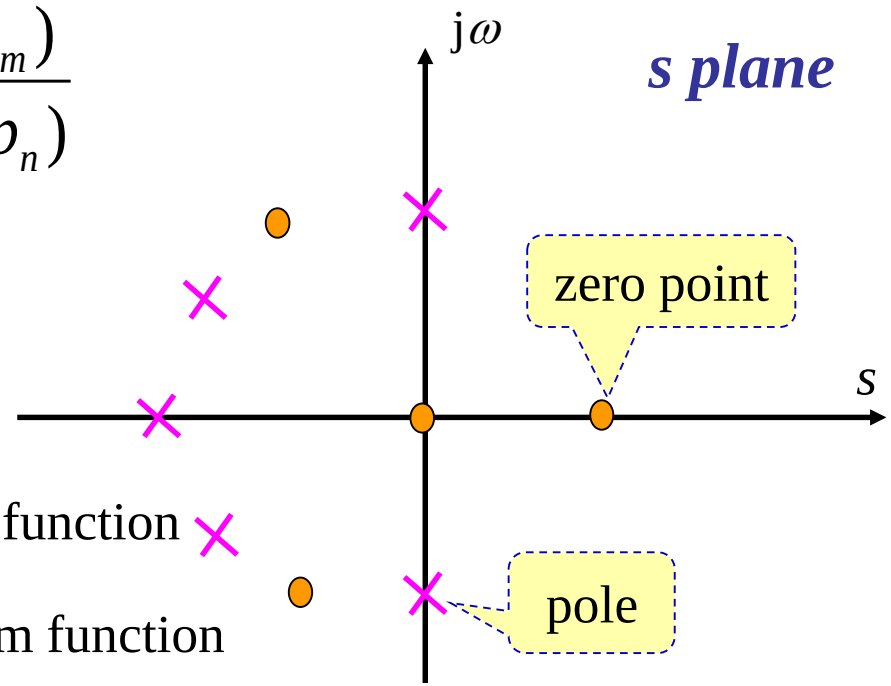
$$H(s)=\frac{b_m s^m+b_{m-1}s^{m-1}+\cdots+b_1 s+b_0}{s^n+a_{n-1}s^{n-1}+\cdots+a_1 s+a_0}=\frac{N(s)}{D(s)}$$

$$=K\frac{(s-z_1)(s-z_2)\cdots(s-z_m)}{(s-p_1)(s-p_2)\cdots(s-p_n)}$$

Zero pole gain form

$z_1, z_2, \cdots, z_m$ is the zero of the system function

$p_1, p_2, \cdots, p_n$ are the poles of the system function

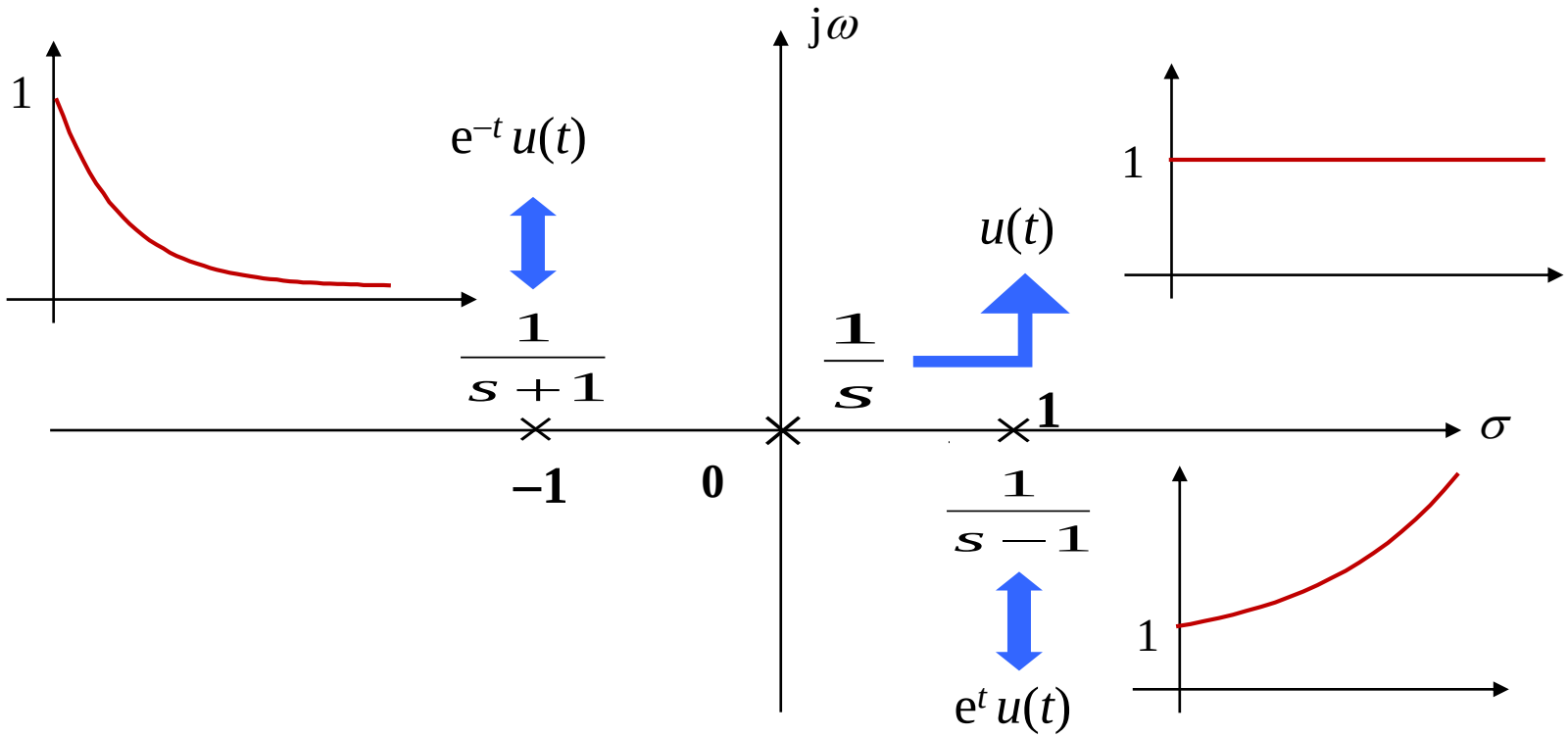


Zero pole distribution diagram

3. System functions and system time domain characteristics

➢ **Relationship between System Function *H*(*s*) and Impulse Response *h*(*t*)**

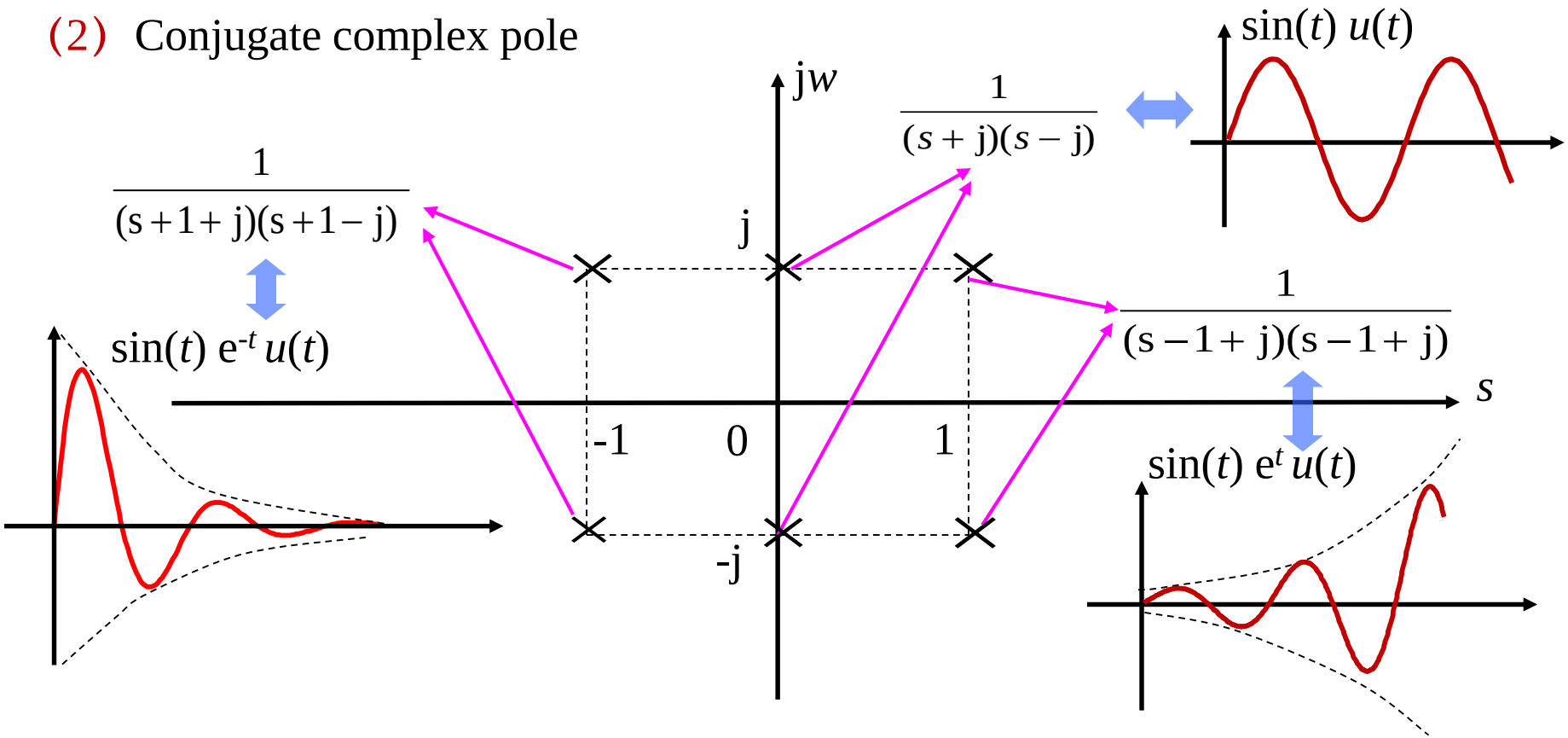
(1) A single pole on the σ axis



3. System functions and system time domain characteristics

➢ **Relationship between system function *H*(*s*) and impulse response *h*(*t*)**

(2) Conjugate complex pole



4. System function and system frequency response

The frequency response $H(\mathrm{j}\omega)$ of the system means that the system is excited by a sinusoidal signal, The steady-state response of the system varies with the frequency of the signal.

For a stable system, $s = \mathrm{j}\omega$ in the system function $H(s)$ to get the system frequency response

$$H(\mathrm{j}\omega) = H(s)\big|_{s=\mathrm{j}\omega}$$

$$H(\mathrm{j}\omega) = \underbrace{|H(\mathrm{j}\omega)|}_{\text{magnitude response}}\mathrm{e}^{\overbrace{\mathrm{j}\phi(\omega)}^{\text{phase response}}}$$

4. System function and system frequency response

System frequency response $H(\mathrm{j}\omega)$

For the system function represented by zero pole gain

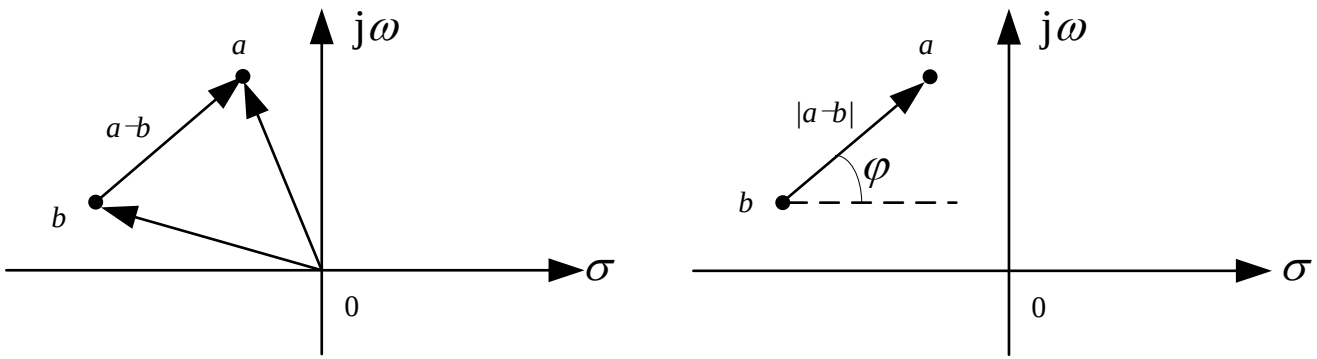
$$H(s) = K\frac{\prod_{j=1}^{m}(s - z_j)}{\prod_{i=1}^{n}(s - p_i)}$$

When the system is stable, let $s=\mathrm{j}\omega$, then

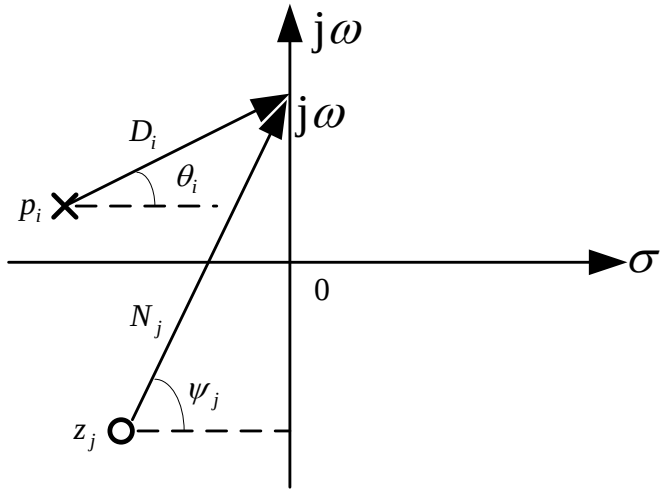
$$H(\mathrm{j}\omega) = K\frac{\prod_{j=1}^{m}(\mathrm{j}\omega - z_j)}{\prod_{i=1}^{n}(\mathrm{j}\omega - p_i)}$$

4. System function and system frequency response

Vector representation of complex numbers *a* and *b* and *a*-*b*



Vector Representation of Zero and Poles of System Function



$$(\mathrm{j}\omega - z_j) = N_j \mathrm{e}^{\mathrm{j}\psi_j}$$

$$(\mathrm{j}\omega - p_i) = D_i \mathrm{e}^{\mathrm{j}\theta_i}$$

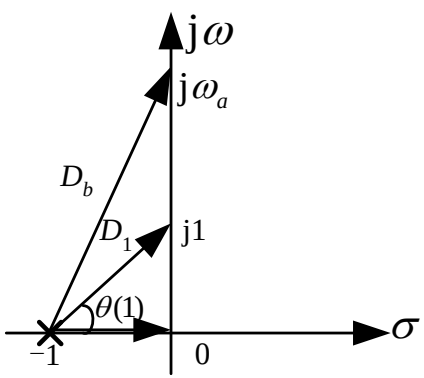
Example: Given the system function , $H(s)=\dfrac{1}{s+1}$ find the frequency response of the system.

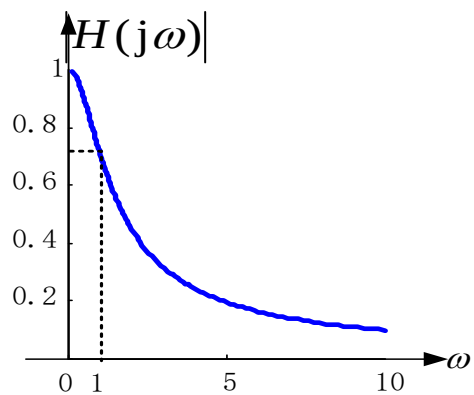
Solution: $H(\mathrm{j}\omega)=H(s)\big|_{s=\mathrm{j}\omega}=\dfrac{1}{\mathrm{j}\omega+1}$

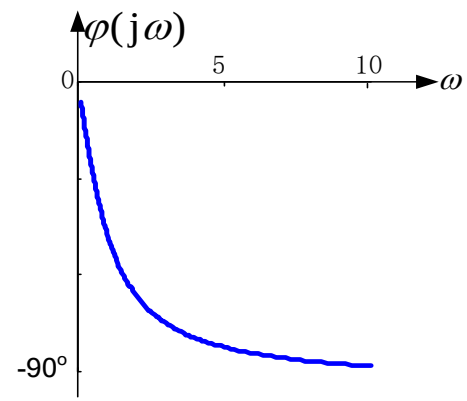
$$|H(\mathrm{j}\omega)|_{\omega=0}=\frac{1}{D_0}=1 \qquad \varphi(\mathrm{j}\omega)|_{\omega=0}=0-\theta_0=0$$

$$|H(\mathrm{j}\omega)|_{\omega=1}=\frac{1}{D_1}=\frac{1}{\sqrt{2}} \qquad \varphi(\mathrm{j}\omega)|_{\omega=1}=0-\theta_1=-\arctan 1=-45^\circ$$

$$|H(\mathrm{j}\omega)|_{\omega\to\infty}=\frac{1}{D_\infty}=0 \qquad \varphi(\mathrm{j}\omega)|_{\omega=0}=0-\theta_\infty=-90^\circ$$



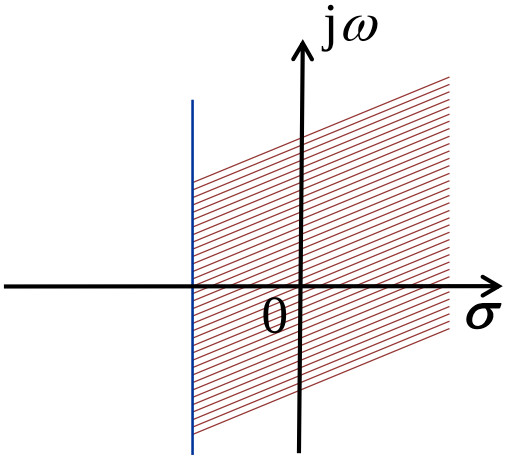




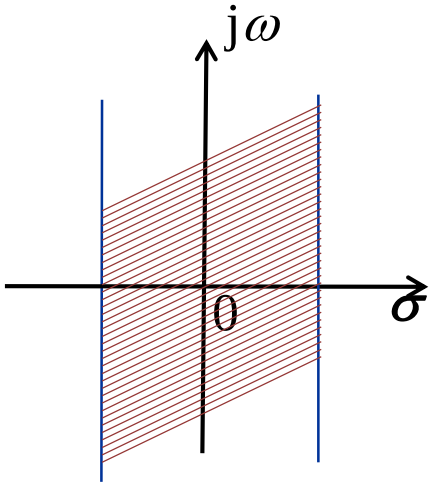
5. System function and system stability

※**A necessary and sufficient condition for the stability of a continuous-time LTI system:**

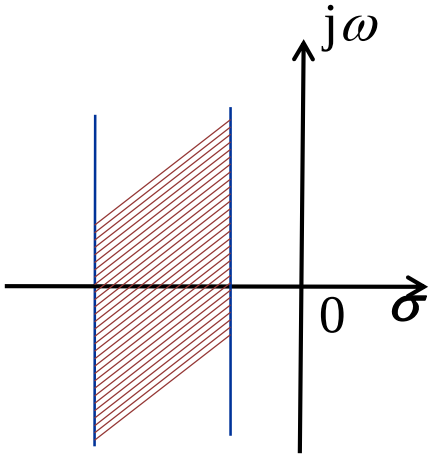
The region of convergence (ROC) of the system function $H(s)$ contains the $j\omega$ axis of the s-plane.



stable system



stable system



unstable system

5. System function and system stability

※Causal continuous LTI system::

The necessary and sufficient condition for the system BIBO to be stable is that all poles of the system function H(s) lie in the left half s-plane.

※ Continuous LTI system:

The necessary and sufficient condition for the system BIBO to be stable is that the **region of convergence (ROC) of the system function H(s) includes the jω axis of the s-plane**.

Example: To judge whether the following causal continuous LTI system is stable.

$$(1)H_1(s)=\frac{s+3}{(s+1)(s+2)},\mathrm{Re}(s)>-1\quad(2)H_2(s)=\frac{s}{s^2+\omega_0^2},\mathrm{Re}(s)>0$$

Solution: Since is causal continuous LTI system

(1) The poles are s= -1 and s= -2, both are in the left half plane of s.

So the causal LTI system is stable.

(2) The pole is $s=\pm j\omega_0$, which is a pair of conjugate poles on the imaginary axis.

So the causal LTI system is unstable.

Comprehensive question: The zero-state response of a continuous-time LTI system is known

$y_{zs}(t) = (0.5 + e^{-t} - 1.5e^{-2t})u(t)$,stimulus signal $x(t)=u(t)$,

Try to find the system function $H(s)$ of the system and draw the zero-pole distribution diagram,Write down the differential equation describing the system, the impulse response $h(t)$of the system,

And judge whether the system is causal and stable.

Solution: The zero-state response and the Laplace transform of the excitation signal are respectively

$$Y_{zs}(s) = \frac{0.5}{s} + \frac{1}{s+1} - \frac{1.5}{s+2} = \frac{2s+1}{s(s+1)(s+2)}, \quad \mathrm{Re}(s) > 0$$

$$X(s) = \frac{1}{s}, \quad \mathrm{Re}(s) > 0$$

According to the definition of the system function, we can get

$$H(s) = \frac{Y_{zs}(s)}{X(s)} = \frac{2s+1}{(s+1)(s+2)} = \frac{2s+1}{s^2+3s+2}, \quad \mathrm{Re}(s) > -1 \quad ①$$

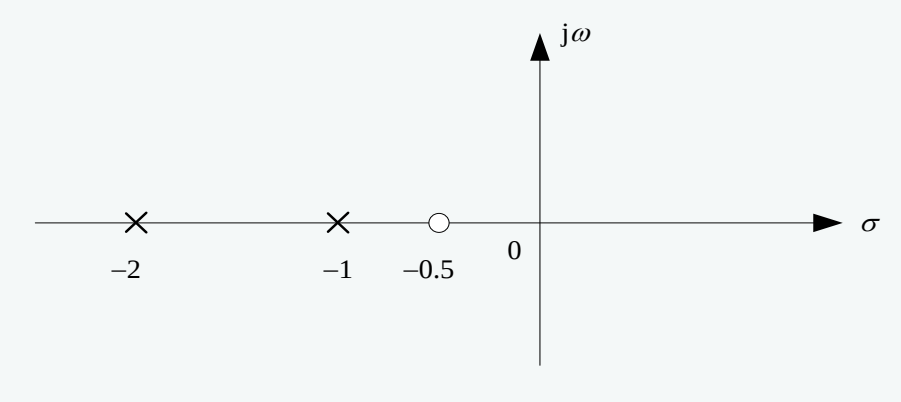
Comprehensive question: The zero-state response of a continuous-time LTI system is known

$y_{zs}(t) = (0.5 + e^{-t} - 1.5e^{-2t})u(t)$,stimulus signal $x(t)=u(t)$,

Try to find the system function $H(s)$ of the system and draw the zero-pole distribution diagram,Write down the differential equation describing the system, the impulse response $h(t)$of the system, And judge whether the system is causal and stable.

Solution: $$H(s) = \frac{Y_{zs}(s)}{X(s)} = \frac{2s+1}{s^2+3s+2}, \quad \mathrm{Re}(s) > -1 \quad ①$$

Zero point of the system function $z = -0.5$, pole is $p_1 = -1$, $p_1 = -2$,



Zero-pole distribution diagram

Comprehensive question: The zero-state response of a continuous-time LTI system is known

$y_{zs}(t) = (0.5 + e^{-t} - 1.5e^{-2t})u(t)$,stimulus signal $x(t)=u(t)$,

Try to find the system function $H(s)$ of the system and draw the zero-pole distribution diagram,Write down the differential equation describing the system, the impulse response $h(t)$of the system, And judge whether the system is causal and stable.

Solution: $$H(s) = \frac{Y_{zs}(s)}{X(s)} = \frac{2s+1}{s^2+3s+2}, \quad \mathrm{Re}(s) > -1 \qquad ①$$

The complex frequency domain expression of the differential equation of the system can be obtained from formula ①

$$(s^2+3s+2)Y_{zs}(s) = (2s+1)X(s)$$

Inverse Laplace transform is performed on both sides, and the differential equation describing the system can be obtained as

$$y''(t) + 3y'(t) + 2y(t) = 2x'(t) + x(t)$$

Comprehensive question: The zero-state response of a continuous-time LTI system is known

$y_{zs}(t) = (0.5 + e^{-t} - 1.5e^{-2t})u(t)$,stimulus signal $x(t)=u(t)$,

Try to find the system function $H(s)$ of the system and draw the zero-pole distribution diagram,Write down the differential equation describing the system, the impulse response $h(t)$of the system,

And judge whether the system is causal and stable.

Solution: Partial fractional expansion of the system function $H(s)$, we can get

$$H(s) = \frac{2s+1}{s^2+3s+2} = \frac{-1}{s+1} + \frac{3}{s+2}$$

Inverse Laplace transform, the impulse response of the system can be obtained as

$$h(t) = (-e^{-t} + 3e^{-2t})u(t)$$

Since the impulse response of this continuous LTI system satisfies

$$h(t) = 0, t < 0$$

Therefore, the continuous LTI system is a causal system

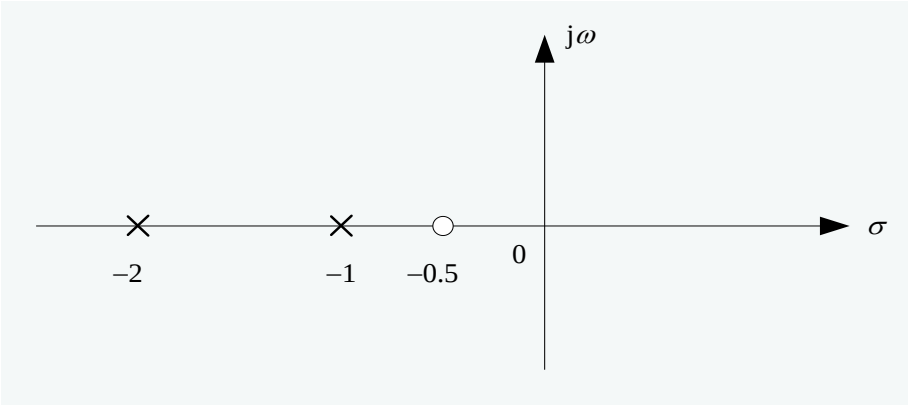
Comprehensive question: The zero-state response of a continuous-time LTI system is known

$y_{zs}(t) = (0.5 + e^{-t} - 1.5e^{-2t})u(t)$,stimulus signal $x(t)=u(t)$,

Try to find the system function $H(s)$ of the system and draw the zero-pole distribution diagram,Write down the differential equation describing the system, the impulse response $h(t)$of the system,

And judge whether the system is causal and stable.

Solution：



For the causal LTI system, it can be seen from the zero-pole distribution diagram that,

The poles of the system are all located in the left half plane of s, **so the system is stable**.

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!